

Research report

Abundance of platelet-derived growth factors (PDGFs), PDGF receptors and activation of mitogen-activated protein kinases in brain decline with age

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Abstract

Platelet-derived growth factors (PDGFs) specifically bind to PDGF receptors (PDGFRs), resulting in their activation via autophosphorylation and subsequent triggering of a cascade of phosphorylation events that include mitogen-activated protein (MAP) kinases. Most of our knowledge concerning MAP kinase activation comes from studies of cultured cells *in vitro*, and little is known about their activation *in vivo*. In the present study, we determined PDGF and PDGFR levels and MAP kinase activities, including extracellular signal-regulated protein kinases (ERK) and c-Jun NH₂-terminal protein kinases (JNK) or stress-activated protein kinases (SAPK) in brain of young and older mice. Both PDGF and PDGFR proteins were most abundant in protein extracts from brain (cerebral cortex) among tissues of heart, liver, spleen, lung and kidney, as determined by Western blot analysis. PDGFR proteins in brain differed significantly between young (1 or 8 weeks) and older (14 months) mice and PDGFR phosphorylation was seen in all age groups examined by a specific antibody against phosphotyrosine. The highest activity of ERK2 was also observed in brain tissues, and this activity declined with age, although ERK1 and ERK2 protein levels were not significantly altered during development and aging. Furthermore, the activity and amount of JNK/SAPK proteins were the most abundant in brain tissues and did not change with age. Thus, our findings demonstrate that the highest levels of PDGFs and PDGFRs existed in brain, and constitutive activation of MAP kinases declined with age, suggesting that signal pathways mediated by PDGF-MAP kinase cascades are important components in coordinating growth and differentiation of neurone and glial cells during development and aging. © 1998 Elsevier Science B.V.

Keywords: PDGF; PDGF receptor; MAP kinase; ERK; JNK; Mouse

1. Introduction

Platelet-derived growth factors (PDGFs) were initially shown to exert potent mitogenic activity on smooth muscle cells [29], and play crucial roles in a variety of physiological and/or pathological processes, such as embryonic development, tissue repair, cell migration, carcinogenesis

and atherosclerosis [3,29,30,36]. PDGFs are composed of the A and B polypeptide chains, which form the three different disulfide-bound dimer proteins PDGF-AA, PDGF-BB, and PDGF-AB [1]. Two different plasma membrane receptors for PDGFs have been identified; receptor α binds all forms of PDGFs, whereas receptor β binds PDGF-BB and, to a lesser extent, PDGF-AB, but not PDGF-AA [24,39]. Each receptor can independently couple with the signalling pathways to evoke a variety of biological responses, including cell proliferation, migration and differentiation. Thus, the presence of specific PDGF isoforms and the differential expression of each PDGF receptor gene product are thought to be major determinants in the spectrum of known cellular responses to PDGFs.

The binding of the ligand dimer induces dimerization of the PDGF receptors (PDGFRs), leading to their activation via autophosphorylation of tyrosine residues in the PDGFR

Abbreviations: AP-1, activator protein-1; ATF, activating transcription factor; ERK, extracellular signal-regulated kinase(s); GST, glutathione S-transferase; JNK, c-Jun NH₂-terminal protein kinase(s); MAPK, mitogen-activated protein kinases; MBP, myelin basic protein; MEK, ERK kinase(s); MKK, MAP kinase kinase(s); NO, nitric oxide; PDGF, platelet-derived growth factor(s); PDGFR, platelet-derived growth factor receptor(s); SAPK, stress-activated protein kinase(s); SEK, SAPK kinase(s).

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kinase domain [31,34]. This triggers a cascade of phosphorylation events involving sequential activation of Ras, Raf, mitogen-activated protein (MAP) kinase kinase, ERK kinase (MEK) and, finally, extracellular regulated-protein kinases (ERKs) (reviewed in [2,4,6,32]). ERKs phosphorylate a variety of regulatory proteins, transcription factors and other protein kinases resulting in the regulation of many effector proteins [6], including 90-kDa ribosomal S6 kinase (p90rsk), cytosolic phospholipase A2, and transcription factors such as Elk-1/TCF, c-fos, ATF2 (a cAMP response element binding protein), NF-IL6 (nuclear factor for the activation of interleukin [6]), and c-Myc [2,4,6,32]. The c-Jun NH₂-terminal kinases (JNK), or stress-activated protein kinases (SAPK), another member of MAP kinase family, are also activated in response to PDGF [7,21]. JNK/SAPK have been demonstrated to activate transcription factors c-Jun and ATF2 [7,13,21]. These transcription factors and effector proteins, mediated by ERKs and JNK/SAPK, play a pivotal role in cell proliferation and differentiation [2,4,6,32].

Each isoform of PDGFs is expressed in the developing brain where they are localized to neurones and growth cones [17]. The PDGF α - and β -receptors are expressed in the mouse forebrain during embryonic and early postnatal development [17]. PDGF α - and β -receptor immunoreactivity has been shown to localize to neurones and glial cells in brains of embryonic and adult mice [17,26,28], although the biological roles of these molecules in the brain remain to be clarified. Furthermore, ERK-mediated phosphorylation of a specific substrate in a physiological context was, in fact, demonstrated first for a neuronal enzyme, tyrosine hydroxylase, the rate-limiting enzyme in catecholamine biosynthesis [15]. However, no systemic investigation of ERK, JNK/SAPK and PDGFR levels has been conducted in multiple tissues of intact animals and no data exist concerning activities of these kinases in brain compared to other organs in development and aging. In the present studies, we performed experiments with protein extracts from various tissues of mice to determine the protein amounts and activities of PDGFs, PDGFRs, ERKs and JNK/SAPK, and show that they are most abundant in brain vs. heart, liver, spleen, lung and kidney. We provide evidence that both PDGF α - and β -receptors are present in the brain in activated or phosphorylated form, as identified by the anti-phosphotyrosine antibody, and also indicated by higher activities of ERK and JNK/SAPK MAP kinases. Our results suggest that the mechanisms controlling ERK and SAPK/JNK activation *in vivo* are influenced by physiological factors that cannot be studied *in vitro*.

2. Materials and methods

2.1. Animals

Female Balb/c mice with aged 1 or 8 weeks, or 14 months were maintained on a light/dark (12/12 h) cycle

at 23°C and received food and water *ad libitum*. All procedures were performed according to protocols approved by the Institution Committee for use and care of laboratory animals. Animals were sacrificed and tissue specimens from heart, liver, spleen, lung, kidney and brain (cerebral cortex) were collected and immediately frozen in liquid nitrogen.

2.2. Western blot analysis

Frozen tissues were homogenized with a polytron homogenizer on ice in buffer containing 20 mM Hepes (pH 7.5), 1.5 mM MgCl₂, 0.2 mM EDTA, 0.1 M NaCl, 0.2 mM DTT, 1 mM Pefablock SC (Boehringer Mannheim, Mannheim, Germany) and 1 μ g/ml leupeptin. The NaCl concentration was adjusted to 0.4 M and cellular debris pelleted by centrifugation for 30 min at 13 000 rpm at 4°C. Supernatants were collected and an equal volume of the same buffer described above containing 0.4 M NaCl and 40% glycerol was added to each tube. The protein concentration was measured by the Bradford Assay (Bio-Rad). Aliquots were frozen in liquid nitrogen and stored at -20°C.

Protein extracts from the tissues were separated by electrophoresis through a 6% (for PDGFRs), 10% (for ERKs and JNK/SAPK) or 15% (for PDGF) SDS-polyacrylamide gel and transferred to an immobilon-p transfer membrane [12]. The membranes were processed with following antibodies, respectively; polyclonal antibodies against PDGF (R and D Systems, Minneapolis, MN) known to react with all three forms of PDGF-AA, PDGF-BB and PDGF-AB; polyclonal antibodies against PDGFR α , PDGFR β and JNK/SAPK (Santa Cruz Biochem., Santa Cruz, CA); anti-phosphotyrosine monoclonal antibody (4G10) (Upstate Biotech. Inc., Lake Placid, NY), and monoclonal antibodies to ERK1 and ERK2 (Transduction Lab., Lexington, KY). Specific antigen-antibody complexes were then detected with the ECL Western Blot Detection Kit (Amersham, Little Chalfont, UK).

2.3. Kinase assay

The procedure used for protein extracts from tissues was similar to that described previously [14,38], with a slight modification. Briefly, frozen tissues were homogenized with polytron homogenizer on ice with buffer A containing 20 mM Hepes (pH 7.4), 50 mM β -glycerophosphate, 2 mM EGTA, 1 mM DTT, 1 mM Na₃VO₄, 1% Triton X-100, 10% glycerol, 2 μ g/ml leupeptin, 400 μ M PMSE, and 1 μ g/ml aprotinin. The homogenate was incubated on ice for 15 min. After centrifugation at 13 000 rpm for 30 min, the supernatant was collected and protein concentration was measured with Bio-Rad protein assay reagent (Bio-Rad).

One half ml of the supernatant containing 0.5 mg proteins was incubated with 8 μ l of antibodies against mammalian JNK/SAPK or ERK2 (Santa Cruz Biochem.)

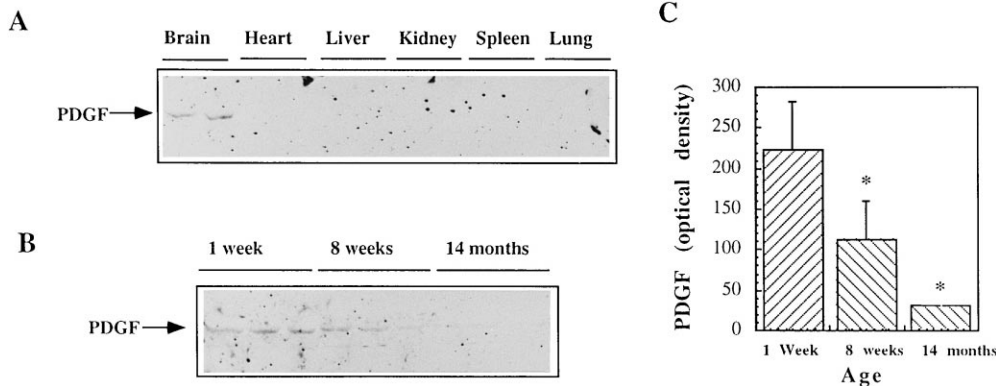


Fig. 1. Western blot analysis of PDGF proteins in tissues. Mice were sacrificed and tissues of heart, lung, liver, brain (cerebral cortex), kidney and spleen were harvested and homogenized. Protein extracts (50 μ g/lane) were separated on 15% SDS-polyacrylamide gel, transferred onto membrane and probed with the antibody against PDGFs, which recognizes all three isoforms of PDGFs. The immunocomplexes were visualised by a Western blot detection kit. Panel A shows PDGF protein levels from a variety of tissues. Panel B presents the results of a representative experiment examining PDGF proteins from the brain of three age groups of mice. Each lane represents an individual animal. Panel C graphically presents PDGF protein levels (means $\pm$ S.D.) from two independent experiments (6 mice/age group). * Significant difference from a group of younger mice. Arrows indicate the positions of PDGFs.

for 2 h at 4°C with rotation. These antibodies have been shown to specifically recognize mammalian p46^{SAPK/JNK} and p54^{SAPK/JNK} or p42^{ERK2} [16,38]. Subsequently, 40 μ l of protein A-Sepharose 4B suspension (Sigma) was added and rotating continued for 1 h at 4°C. The immunocomplexes were precipitated by centrifugation and washed 2 times with buffer A, B (500 mM LiCl, 100 mM Tris, 1 mM DTT, 0.1% Triton X-100; pH 7.6), and C (20 mM Mops, 2 mM EGTA, 10 mM MgCl₂, 1 mM DTT, 0.1% Triton X-100; pH 7.2), respectively.

ERK2 activities in the immunocomplexes were measured as described previously [14,16,23,38]. Briefly, immunocomplexes were incubated with 35 μ l of buffer C supplemented with myelin basic proteins (MBP; Upstate Biotech. Inc.) (5 μ g), [γ -³²P]ATP (Amersham, Little Chalfont, UK), and MgCl₂ (50 mM) for 20 min at 37°C with vortexing every 5 min. To stop the reaction, 15 μ l of 5 $\times$ Laemmli buffer was added and the mixture was boiled for 5 min. Proteins in the kinase reaction were resolved by SDS-PAGE (12% gel) and subjected to autoradiography.

The assay for JNK/SAPK activity was performed similarly as described above. The substrate used was the glutathione S-transferase (GST)-c-Jun fusion proteins (the plasmid was provided by Dr. Woodgett), which was produced in *Escherichia coli* and isolated using glutathione Sepharose 4B RediPack Columns (Pharmacia Biotech Inc., Piscataway, NJ) according to the manufacturer's protocol. Proteins in the kinase reaction were resolved by SDS-PAGE (12% gel) and subjected to autoradiography [14,16,23,38].

2.4. Statistical analysis

Analysis of variance was performed for multiple comparisons. The Mann-Whitney *U* test was used for comparison between two groups. A *P* value less than 0.05 was considered statistically significant.

3. Results

3.1. PDGF and PDGFR in brain

To determine protein levels of PDGFs in tissues, 8-week old Balb/c mice were sacrificed and various tissues were

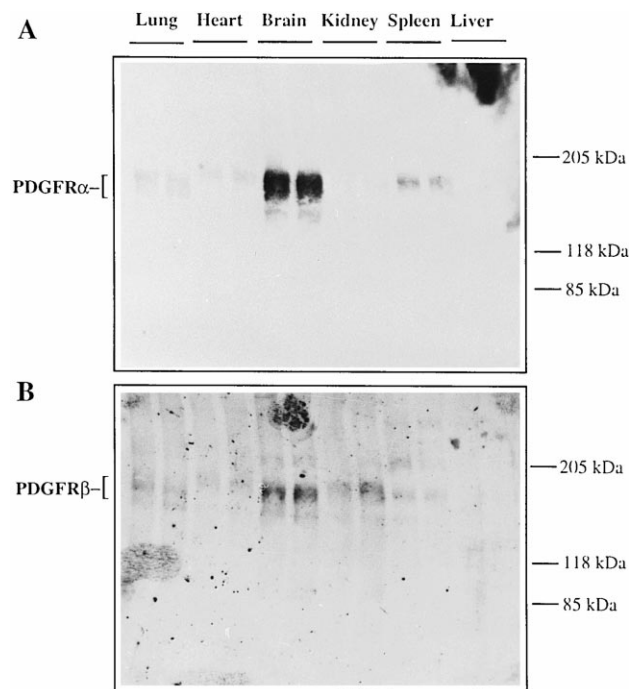


Fig. 2. Western blot analyses of PDGFRs in tissues of 8-week-old mice. Tissue samples from heart, lung, kidney, liver, spleen and brain (cerebral cortex) were harvested and homogenized. Protein extracts (50 μ g/lane) from tissues were separated on 6% SDS-polyacrylamide gel, transferred onto membrane and probed with the antibody against PDGFR α (A) or PDGFR β (B). The immunocomplexes were visualised by a Western blot detection kit. Each lane represents an individual animal. Data show results of a representative experiment. Lines on right side indicate molecular weights.

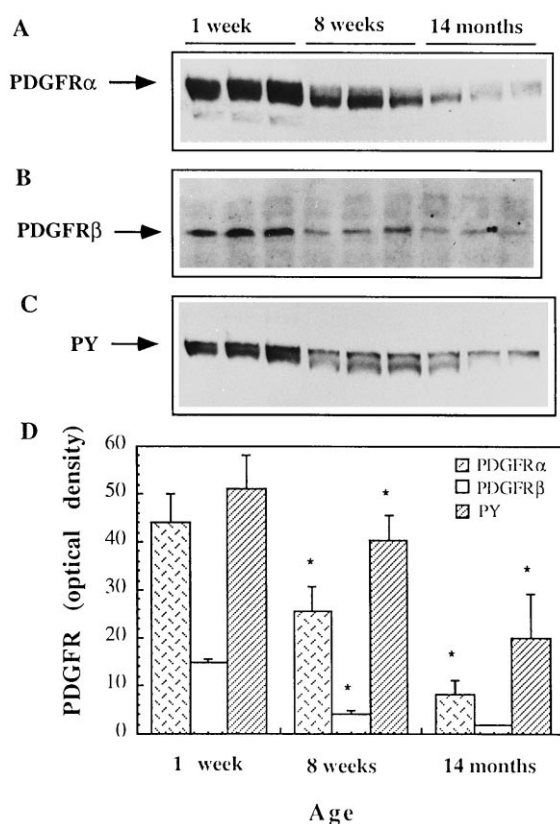


Fig. 3. Effects of age on PDGFRs in brain. Protein extracts (50 μ g/lane) from brain tissues were separated on 6% SDS-polyacrylamide gel, transferred onto membrane and probed with the antibody against PDGFR α (A), PDGFR β (B), phosphotyrosine (C; PY). The immunocomplexes were visualised by a Western blot detection kit. Panels A–C show results of a representative experiment. Each lane represents an individual animal. Panel D graphically presents protein levels (mean $\pm$ S.D.) from two independent experiments. * Significant difference from a group of younger animals ($n = 6$).

harvested and homogenized. Protein extracts were analyzed by Western blot using specific antibodies to all three isoforms of PDGF. Data shown in Fig. 1A provided evidence that PDGFs were detectable in brain tissues (cerebral cortex) but not heart, kidney, brain, spleen or lung. PDGFs of brain tissue protein extracts from the three groups were determined by Western blot analysis and compared. Fig. 1B shows an age-dependent decrease of PDGF proteins in brain, while PDGF proteins from 14-month-old mice were almost undetectable (Fig. 1B).

It has been shown that two forms of PDGFRs exist in mammalian cells: mature PDGFR with 180 kDa, and immature, with 160 kDa [8]. Our results demonstrated that the major band of PDGFRs was of the mature form (about 180 kDa) and a weaker band represented the immature form (about 160 kDa) (Fig. 2). Surprisingly, the most abundant PDGFRs were found in brain tissue, although higher levels of PDGFRs were seen in protein extracts from lung and spleen, and immunoreactivity of PDGFRs was undetectable in liver tissue (Fig. 2). The proportion of PDGFR α is much higher than PDGFR β in brain tissue. No other bands were seen on the blots verifying the specificity of both antibodies.

In the developing central nervous system, PDGFRs are constitutively produced and are crucial in regulating cell proliferation and differentiation of oligodendrocyte precursors before myelination [40]. To determine whether PDGFRs are also present and activated in adult and aged mice, we performed experiments with protein extracts from brain tissues of mice at 1 week, 8 weeks and 14 months of age. As shown in Fig. 3A, the amount of PDGFR α proteins was highest in brain in all three groups, and declined in an age-dependent manner. A similar observation was seen in PDGFR β protein levels (Fig. 3B). To

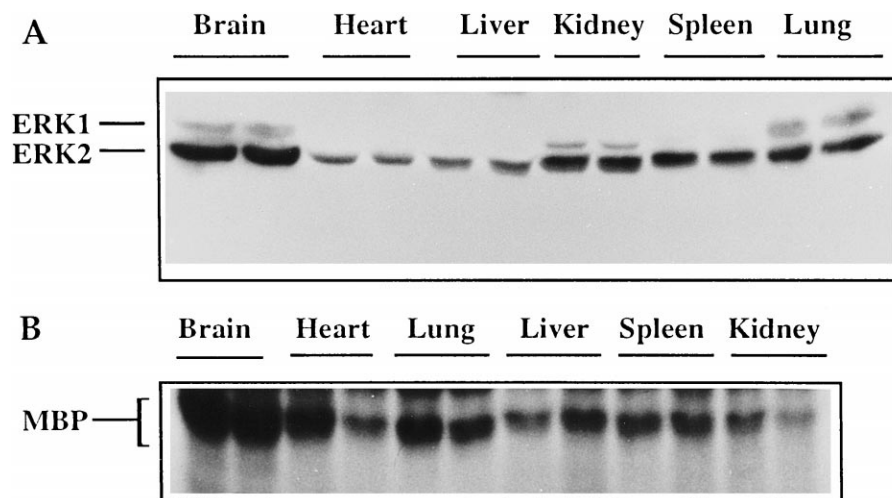


Fig. 4. Analysis of ERK in mouse tissues. Mice (8 weeks) were sacrificed and tissue samples from heart, lung, liver, spleen, brain and kidney was harvested and homogenised. Panel A represents results of Western blot analysis for ERK1 and ERK2. Protein extracts (50 μ g/lane) from tissues were separated on 10% SDS-polyacrylamide gel, transferred onto membrane and probed with the antibody against ERK1 and ERK2. The immunocomplexes were visualised by a Western blot detection kit. Panel B shows ERK2 kinase activity. P42^{ERK2} proteins were immunoprecipitated from the protein extracts (0.5 mg in 0.5 ml of kinase buffer A) and their kinase activities measured based on phosphorylation of myelin basic protein (MBP). Each lane represents an individual animal.

address the question of whether these PDGFRs are activated or phosphorylated *in vivo*, PDGFR phosphorylation was assessed using a specific monoclonal antibody against phosphotyrosine. Fig. 3C shows that PDGFRs were phosphorylated in proteins extracted from brain of all three age groups at an intensity that declined with age, supporting the notion that PDGFRs are in an activated state *in vivo*. Fig. 3D summarizes data showing PDGFR α and PDGFR β levels and phosphotyrosine as determined by quantification of optical densities from autoradiograms of two experiments. Significant differences were found in PDGFRs and phosphotyrosine of brain tissues between 1 week and 14 months of age, but not in PDGFR β between 8 weeks and 14 months. In addition, PDGFR α and phosphotyrosine protein levels were also markedly decreased in brain of 14-month-old vs. 8-week-old mice (Fig. 3D).

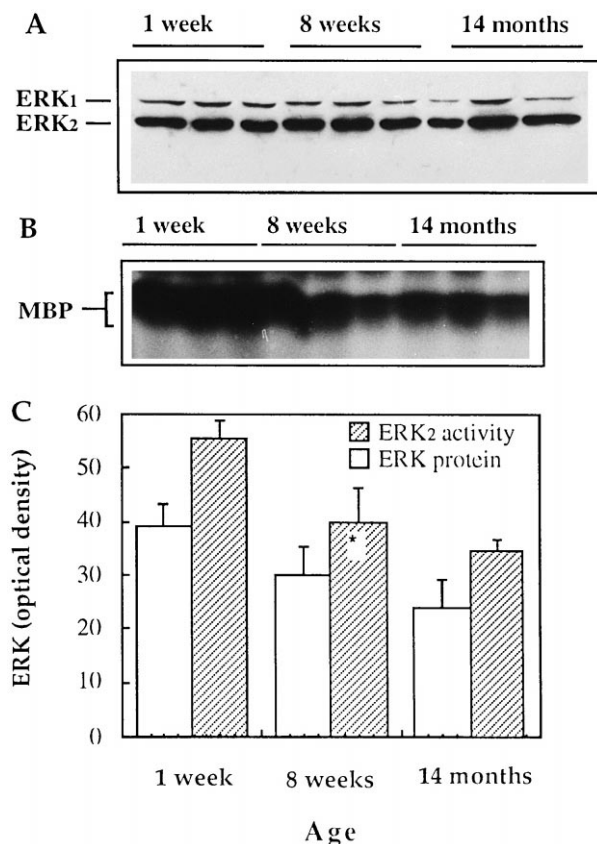


Fig. 5. Effects of age on ERKs in brain. Panel A: Western blot analysis of ERKs. Protein extracts (50 μ g/lane) from brain tissues were separated on 10% SDS-polyacrylamide gel, transferred onto membrane and probed with the antibody against ERK1 and ERK2. The immunocomplexes were visualised by a Western blot detection kit. Panels B shows the results of a representative experiment with ERK kinase activities. P42^{ERK2} proteins were immunoprecipitated from the protein extracts (0.1 mg in 0.5 ml of kinase buffer A) and their kinase activities measured based on phosphorylation of myelin basic protein (MBP). Each lane represents an individual animal. Panel C graphically presents proteins and activities of ERKs (mean $\pm$ S.D.) from two independent experiments. * Significant difference from a group of younger animals; $n = 6$.

3.2. Age-dependent decline in ERK activity in brain

Binding of several neurotrophic factors, including PDGF, to their receptors rapidly activates both ERKs and JNK/SAPK in a variety of cells *in vitro* [2,4,6,31,32,34]. As shown above, PDGFRs were phosphorylated or activated in brain, and we hypothesized that would be possible to detect activation of ERKs in brain tissue *in vivo*. To verify this hypothesis, protein extracts from brain, liver, heart, spleen, lung and kidney were analyzed for ERK activity determined using MBP as a substrate, and ERK2 activities were found to be the highest in brain tissues (Fig. 4B). In addition, ERK1 and ERK2 protein levels in brain tissue appeared much higher than in other tissues (Fig. 4A). Interestingly, ERK1 was not detectable in heart, liver and spleen, suggesting a tissue specific expression of ERK1.

ERK activities showed an age-dependent decline concomitant with the observation for PDGFRs. When ERK2 was isolated by immunoprecipitation from proteins extracted from brain tissues of three age groups, ERK2 activities decreased significantly with age (Fig. 5B). However, ERK1 and ERK2 protein levels were not significantly altered during aging, as evidenced by Western blot analysis (Fig. 5A). These data confirm that ERK is not regulated at the level of translation *in vivo*. Fig. 5C summarizes the data showing ERK2 activation and protein levels of ERK1 and ERK2, as determined by quantification of optical densities from autoradiograms of two experiments. A significant difference in ERK2 activities was seen among the three groups, but not ERK1 and ERK2 protein levels.

3.3. Abundant SAPK in brain tissue

Both JNK/SAPK and ERKs can be activated by a variety of stimuli, including cellular stresses, cytokines and growth factors *in vitro* [2,4,6,32,34]. However, the activation of kinases vary depending on the stimulating signals. It has been established that JNK/SAPK phosphorylate Ser-63 and Ser-73 of the NH₂-terminal transactivation domain of c-Jun *in vitro* may be responsible for phosphorylation of c-Jun *in vivo* [18]. We used GST-c-Jun (1–135) as a substrate in a kinase assay to identify the protein kinase that binds to and phosphorylates the N-terminal domain of c-Jun. We analyzed the activity and amount of JNK/SAPK in various types of tissues of mice. Fig. 6 shows activities and amounts of JNK/SAPK in protein extracts of brain, heart, liver, spleen, lung and kidney as determined by the kinase assay and Western blot analysis. As seen in Fig. 6B the highest JNK/SAPK activities were observed in brain tissue at levels 50–100 times higher than in liver tissues (Fig. 6B), and JNK/SAPK proteins also appeared much more abundant in protein extracts from brain (Fig. 6A). No other bands were seen in the blot,

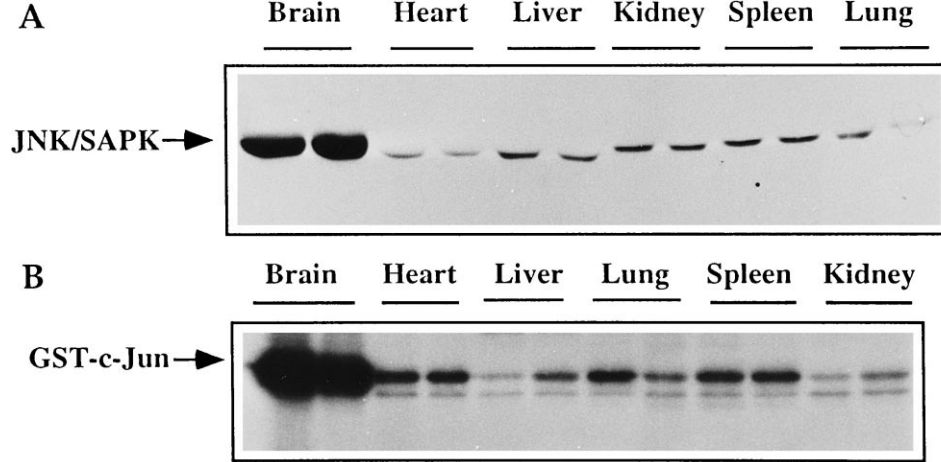


Fig. 6. Analysis of JNK/SAPK in mouse tissues. Mice (8 weeks) were sacrificed and tissue samples from heart, lung, liver, spleen, brain and kidney were harvested and homogenised. Panel A shows results of Western blot analysis for JNK/SAPK. Protein extracts (50 μ g/lane) from tissues were separated on 10% SDS-polyacrylamide gel, transferred onto membrane and probed with the antibody against JNK/SAPK. The immunocomplexes were visualised by a Western blot detection kit. Panel B shows JNK/SAPK kinase activity. P46,52^{JNK1/SAPK} proteins were immunoprecipitated from the protein extracts (0.1 mg in 0.5 ml of kinase buffer A) and their kinase activities were measured based on phosphorylation of GST-c-Jun. Each lane represents an individual animal. Data show a representative blot from four independent experiments with similar results.

verifying the specificity of the JNK/SAPK antibody used in this study. This phenomena does not occur in cultured cells, since no significant difference in JNK/SAPK activity and protein levels was found between untreated fibroblasts and PC12 cells derived from brain tissue (data not shown).

The observation of an age-dependent decline in ERK2 activities in brain (Fig. 5) led us to simultaneously determined JNK/SAPK activities in protein extracts from tissues of mice in the three-age groups. JNK/SAPK activities showed no significant difference among all tissues of the three groups of mice (Fig. 7B), and JNK/SAPK

protein level were also not significantly changed during aging (Fig. 7A). The activities and protein levels of JNK/SAPK were much higher in brain even in older mice, providing the first evidence of abundant JNK/SAPK in brain tissue. Any additional role of JNK/SAPK in brain function has yet to be proven.

4. Discussion

In the present study, we demonstrated that PDGFR α and PDGFR β proteins were constitutively produced and,

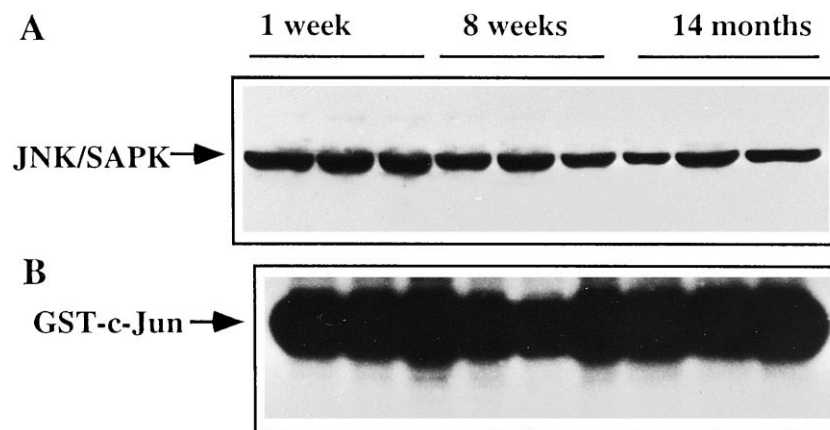


Fig. 7. JNK/SAPK in brain of young and older mice. Panel A: Western blot analysis of JNK/SAPK. Protein extracts (50 μ g/lane) from brain tissues were separated on 10% SDS-polyacrylamide gel, transferred onto membrane and probed with the antibody against JNK/SAPK. The immunocomplexes were visualised by a Western blot detection kit. Panels B shows a representative experiment with JNK/SAPK kinase activities. P46,52^{JNK1/SAPK} proteins were immunoprecipitated from the protein extracts (0.1 mg in 0.5 ml of kinase buffer A) and their kinase activities were measured based on phosphorylation of GST-c-Jun. Each lane represents an individual animal. Note no significant difference between young and older mice.

at least in part, presented in an activated state, i.e. phosphorylated forms, in the cerebra of young and older mice. In view of the wide distribution of PDGFRs in mature neurones and glial cells and the broad functions in cell proliferation, migration and differentiation (reviewed in [1,3,24,30,34,36]), we believe PDGFRs are important components of the signaling network in the brain that coordinates interaction, growth and differentiation of these cells during development and aging. On the other hand, it is unknown whether alterations in PDGFRs and MAP kinase activity reflect changes during aging in individual cells or changes among different cell types, i.e. neurones and/or glial cells in brain. Further study in this issue should enhance our knowledge for understanding the role of PDGFR-MAP kinase signal pathways in brain aging.

The precise functions of PDGFRs in brain *in vivo* remains to be elucidated. However, studies using cultured neurones *in vitro* have provided evidence that neurotrophic, instructive, and neuroprotective effects of PDGFs on neurones were mediated through the activation of the two forms of PDGFRs [25]. For instance, PDGF-BB exerted trophic activity on cultured GABAergic interneurons by increasing the expression of glutamic acid decarboxylase and the survival of these cells [33]. PDGFs promoted survival of rat and human mesencephalic dopaminergic neurones in culture [27], and, *in vitro*, induced neurototic outgrowth of the PC12 neuronal cell line [9]. Recently, Cheng and Mattson [5] also reported that PDGFs protect embryonic hippocampal neurones against energy deprivation and oxidative injury *in vitro*, by increasing the activity of antioxidative enzymes such as catalase and glutathione peroxidase. The expression of both PDGFRs in brain and the fact that both PDGF-AA and PDGF-BB effectively protected these cells, strongly suggest that either α -receptors or β -receptors can trigger neuroprotective mechanisms. Our findings of predominant PDGF-PDGFR proteins in brain, together with observations by others, support the notion that the production of PDGFs and their receptors in mature neurones and glial cells could have a physiological role in maintaining nerve system function *in vivo*.

Binding of PDGF to their receptors causes activation of MAP kinase cascades in cultured cells, resulting in cell proliferation. There is evidence that glial cells and even neurones in some areas of the brain show age-dependent proliferation in the central nervous system of adult mice [20]. We demonstrated that the abundant PDGFR proteins coincided with higher activities of both MAP kinases, ERK2 and JNK/SAPK, in the cerebra, supporting the role of MAP kinases in cell growth *in vivo*. As described above, ERK and JNK/SAPK kinases can phosphorylate or activate several transcription factors, including c-Fos, c-Jun, ATF₂, Elk-1 [11,19]. c-Fos and c-Jun dimerize to form the transcription factor AP-1, which binds to AP-1 consensus sequences present in numerous genes associated with cell proliferative response and extracellular matrix

production, such as FGF, endothelin-1, *c-myc*, collagenase and TGF- β 1 [11,19]. ATF₂ can dimerize not only with c-Jun, but also with itself and some other members of the ATF family, including ATF₃, CREB and the closely-related NF- κ B. Elk-1, together with serum response factor, controls transcription from the serum response element. These transcription factors regulate gene expression, including matrix metalloproteinase, adhesion molecule E-selectin, NO synthase, IL-8 and proliferating cell nuclear antigen [11,22]. These genes have been demonstrated to play a key role in cell growth and cellular homeostasis. Thus, our observations of ERK and JNK/SAPK activation in brain could be important to our understanding of the mechanism controlling expression of these genes during development and aging.

Recent studies have provided evidence that activation of MEKK, a JNK/SAPK kinase, and JNK/SAPK may be involved in the process leading to cell apoptosis [35,37]. Evidence also indicates that apoptosis naturally occurred in developing brain of rats [10]. Given the higher activation of JNK/SAPK in brain tissues of animals including older mice, we suggest that JNK/SAPK plays a role in mediating apoptosis *in vivo* during development and aging. We found significant decreases of PDGFRs and ERKs, but not JNK/SAPK in older mice. The balance of activities between ERK and JNK/SAPK may be a determinant of growth or apoptosis in brain during aging, e.g., the predominant activation of JNK/SAPK compared to ERK may impair the balance between cell growth and apoptosis in brain during aging. Therefore, further study of signaling pathways mediated by ERKs and JNK/SAPK in brain may provide valuable information for understanding the mechanism of brain development and aging.

Acknowledgements

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References

- [1] H.N. Antoniades, Human platelet-derived growth factor: purification of PDGF-I and PDGF-II and separation of their reduced subunits, *Proc. Natl. Acad. Sci. USA* 78 (1981) 7314–7317.
- [2] J. Avruch, X.-F. Zhang, J.M. Kyriakis, Raf meets Ras: completing the framework of a signal transduction pathway, *Trends Biochem. Sci.* 19 (1994) 279–283.

- [3] F. Besnard, F. Perraud, M. Sensenbrenner, G. Labourette, Platelet-derived growth factor is a potent mitogen for glial cell but not neuronal rat brain cells in vitro, *Neurosci. Lett.* 73 (1987) 287–292.
- [4] E. Cano, L.C. Mahadevan, Parallel signal processing among mammalian MAPKs, *Trends Biochem. Sci.* 20 (1995) 117–122.
- [5] B. Cheng, M. Mattson, PDGFs protect hippocampal neurons against energy deprivation and oxidative injury: evidence for induction of antioxidant pathways, *J. Neurosci.* 15 (1995) 7095–7104.
- [6] R.J. Davis, The mitogen-activated protein kinase signal transduction pathway, *J. Biol. Chem.* 268 (1993) 14553–14556.
- [7] B. Dérjard, M. Hibi, I.-H. Wu, T. Barrett, B. Su, T. Deng, M. Karin, R.J. Davis, JNK1: a protein kinase stimulated by UV light and Ha-Ras that binds and phosphorylates the c-Jun activation domain, *Cell* 76 (1994) 1025–1037.
- [8] A. Eriksson, A. Siegbahn, B. Westermark, C.H. Heldin, L. Claesson-Welsh, PDGF α - and β -receptors active unique and common signal transduction pathways, *EMBO J.* 11 (1992) 543–550.
- [9] G.R. Fanger, J.R. Jones, R.A. Maue, Differential regulation of neuronal sodium channel expression by endogenous and exogenous tyrosine receptors expressed in rat pheochromocytoma cells, *J. Neurosci.* 15 (1995) 201–213.
- [10] I. Ferrer, M. Olive, J. Ribera, A.M. Planas, Naturally occurring (programmed) and radiation-induced apoptosis are associated with selective-Jun expression in the developing rat brain, *Eur. J. Neurosci.* 8 (1996) 1286–1298.
- [11] T. Force, C.M. Pombo, J.A. Avruch, J.V. Bonventre, J.M. Kyriakis, Stress-activated protein kinases in cardiovascular disease, *Circ. Res.* 78 (1996) 947–953.
- [12] M. Gorospe, Y. Liu, Q. Xu, F.J. Chrest, N.J. Holbrook, Inhibition of G_1 cyclin-dependent kinase activity during growth arrest of human breast carcinoma cells by prostaglandin A_2 , *Mol. Cell. Biol.* 16 (1996) 762–770.
- [13] S. Gupta, D. Campbell, B. Dérjard, R.J. Davis, Transcription factor ATF2 regulation by the JNK signal transduction pathway, *Science* 267 (1995) 389–393.
- [14] K.Z. Guyton, Y. Liu, M. Gorospe, Q. Xu, N.J. Holbrook, Activation of mitogen-activated protein kinases by H_2O_2 : role in cell survival following oxidant injury, *J. Biol. Chem.* 271 (1996) 4138–4142.
- [15] J.W. Haycock, N.G. Ahn, M.H. Cobb, E.G. Krebs, ERK1 and ERK2, two microtubule-associated protein 2 kinases, mediate phosphorylation of tyrosine hydroxylase at serine-31 in situ, *Proc. Natl. Acad. Sci. USA* 89 (1992) 2365–2369.
- [16] Y. Hu, B. Metzler, Q. Xu, Discordant activation of stress-activated protein kinases or c-Jun NH₂-terminal protein kinases in tissues of heat-stressed mice, *J. Biol. Chem.* 272 (1997) 9113–9119.
- [17] J.B. Hutchins, V.E. Jefferson, Developmental distribution of platelet-derived growth factor in the mouse central nervous system, *Dev. Brain Res.* 67 (1992) 121–135.
- [18] T. Kallunki, B. Su, I. Tsigelny, H.K. Sluss, B. Derjard, G. Moore, R.J. Davis, M. Karin, *Genes Dev.* 8 (1994) 2996–3007.
- [19] M. Karin, The regulation of AP-1 activity by mitogen-activated protein kinases, *J. Biol. Chem.* 270 (1995) 16483–16486.
- [20] H.G. Kuhn, H. Dickinson-Anson, F.H. Gage, Neurogenesis in the dentate gyrus of the adult rat: age-related decrease of neuronal progenitor proliferation, *J. Neurosci.* 16 (1996) 2027–2033.
- [21] J.M. Kyriakis, P. Banerjee, E. Nikolakaki, T. Dai, E.A. Rubie, M.F. Ahmad, J. Avruch, J.D. Woodgett, The stress-activated protein kinase subfamily of c-Jun kinases, *Nature* 369 (1994) 156–160.
- [22] J.M. Kyriakis, J. Avruch, Sounding the alarm: protein kinase cascades activated by stress and inflammation, *J. Biol. Chem.* 271 (1996) 24313–24316.
- [23] Y. Liu, K.Z. Guyton, M. Gorospe, Q. Xu, G. Kokkonen, Y. Mock, G.S. Roth, N.J. Holbrook, Age-related decline in mitogen-activated protein kinase activity in epidermal growth factor-stimulated rat hepatocytes, *J. Biol. Chem.* 271 (1996) 3604–3607.
- [24] T. Matsui, M. Heidaran, T. Miki, N. Popescu, W. Larochelle, M. Kraus, J. Pierce, J. Aaronson, Isolation of a novel receptor cDNA establishes the existence of two PDGF receptor genes, *Science* 243 (1989) 800–804.
- [25] T. Matsui, K. Sano, T. Tsukamoto, M. Ito, T. Takaishi, Human neuroblastoma cells express α and β platelet-derived growth factor receptors coupling with neurotrophic and chemotactic signaling, *J. Clin. Invest.* 92 (1993) 1153–1160.
- [26] M. Mercola, C. Wang, J. Kelly, C. Brownlee, L. Jackson-Grusby, C. Stiles, D.F. Bowen-pope, Selective expression of PDGF A and its receptor during early mouse embryogenesis, *Dev. Biol.* 138 (1990) 114–122.
- [27] G. Nikkha, P. Odin, A. Smits, A. Tingström, A. Othberg, P. Brundin, K. Funa, O. Lindvall, Platelet-derived growth factor promotes survival of rat and human mesencephalic dopaminergic neurons in culture, *Exp. Brain Res.* 92 (1993) 516–523.
- [28] N.B. Oumesmar, L. Vignais, A. Baron-van Evercooren, Developmental expression of platelet-derived growth factor α -receptor in neurons and glial cells of the mouse CNS, *J. Neurosci.* 17 (1997) 125–139.
- [29] R. Ross, J. Glomet, B. Kariya, L. Harker, A platelet-dependent serum factor that stimulates the proliferation of arterial smooth muscle cells in vitro, *Proc. Natl. Acad. Sci. USA* 71 (1974) 1207–1210.
- [30] R. Ross, The pathogenesis of atherosclerosis: a perspective for the 1990s, *Nature* 362 (1990) 801–809.
- [31] M. Sasahara, J.W.U. Fries, E.W. Raines, A.M. Grown, L.E. Westrum, M.P. Frosch, D.T. Bonthron, R. Ross, T. Collins, PDGF- β chains in neurons of the central nervous system, posterior pituitary and in transgenic model, *Cell* 64 (1991) 217–227.
- [32] R. Seger, E.G. Krebs, The MAPK signaling cascade, *FASEB J.* 9 (1995) 726–735.
- [33] A. Smits, E. Ballagi, K. Funa, PDGF-BB exerts trophic activity on cultured GABA interneurons from the newborn rat cerebellum, *Eur. J. Neurosci.* 5 (1993) 986–994.
- [34] A. Ullrich, J. Schlessinger, Signal transduction by receptor with tyrosine kinase activity, *Cell* 61 (1990) 203–212.
- [35] M. Verheij, R. Bose, X.H. Lin, Z. Fuks, R.N. Kolesnick, Requirement for ceramide-initiated SAPK/JNK signalling in stress-induced apoptosis, *Nature* 380 (1996) 75–79.
- [36] B. Westermark, A. Wasteson, A platelet factor stimulating human normal glial cells, *Exp. Cell Res.* 98 (1976) 170–174.
- [37] Z. Xia, M. Dickens, J. Raingeaud, R.J. Davis, M.E. Greenberg, Opposing effects of ERK and JNK-p38 MAP kinases on apoptosis, *Science* 270 (1995) 1326–1331.
- [38] Q. Xu, Y. Liu, M. Gorospe, R. Udelsman, N.J. Holbrook, Acute hypertension activates mitogen-activated protein kinases in arterial wall, *J. Clin. Invest.* 97 (1996) 508–514.
- [39] Y. Yarden, A. Ullrich, Growth factor receptor tyrosine kinases, *Annu. Rev. Biochem.* 54 (1988) 443–478.
- [40] H.J. Yeh, K.G. Ruit, Y.X. Wang, W.C. Parks, W.D. Snider, T.F. Deuel, PDGF- α chain gene is expressed by mammalian neurons during development and in maturity, *Cell* 84 (1991) 209–216.